

Truth Table

Design a decoder to decode letter n and the last three digits of your 900 number and display decimal numbers in a seven-segment decoder.

- A. Write the Truth Table for the circuit for all possible inputs with the corresponding outputs.
- B. Draw K-maps for each segment 'a' to 'g' and obtain a simplified logic combination of inputs for each output.

X	Y	Z	Display	a	b	c	d	e	f	g
0	0	0	n	0	0	1	0	1	0	1
0	0	1	6	1	0	1	1	1	1	1
0	1	0	9	1	1	1	0	1	1	1
0	1	1	3	1	1	1	0	0	1	1
1	0	0	-	0	0	0	0	0	0	1
1	0	1	-	0	0	0	0	0	0	1
1	1	0	-	0	0	0	0	0	0	1
1	1	1	-	0	0	0	0	0	0	1

K-Maps

Segment A	Z	Z'
X'Y'	1	0
X'Y	1	1
XY	0	0
XY'	0	0

Segment B	Z	Z'
X'Y'	0	0
X'Y	1	1
XY	0	0
XY'	0	0

Segment C	Z	Z'
X'Y'	1	1
X'Y	1	1
XY	0	0
XY'	0	0

Segment D	Z	Z'
X'Y'	1	0
X'Y	0	0
XY	0	0
XY'	0	0

Segment E	Z	Z'
X'Y'	1	1
X'Y	0	1
XY	0	0
XY'	0	0

Segment F	Z	Z'
X'Y'	1	0
X'Y	1	1
XY	0	0
XY'	0	0

Segment G	Z	Z'
X'Y'	1	1
X'Y	1	1
XY	1	1
XY'	1	1